

2023-Ph-H1-Q9

Section: Electricity

Topic: Current, PD, Power, Resistance

Which statement is true when a resistor obeys Ohm's law?

Answer: A

(Current is directly proportional to voltage)

Guidance for Students:

Ohm's Law: $V = IR$ implies that if resistance is constant, then current increases **linearly** with voltage.

Revision Tips:

- Ohmic devices: straight-line V - I graph through origin.
- Non-ohmic: filament lamp, diode, etc.
- Check units: V (volts), I (amps), R (ohms)